

# ALBERT JOSHWA A

Java Backend Engineer | Microservices & Distributed Systems | AI / LLM Systems

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## PROFESSIONAL SUMMARY

Java backend engineer with 4+ years building enterprise SaaS at Zoho Corporation and GoFrugal Technologies (Zoho group), serving customers in 150+ countries. Led the monolith-to-microservices decomposition of Zoho Books, splitting 15+ tightly coupled APIs into independently deployable services now shared across 6+ Zoho Finance products, and shipped a real-time bidirectional sync engine live with paying customers on the ONDC Network after clearing internal security and DPIA review. Also ships production systems end to end independently, including a live payments platform on Cloudflare Workers and D1 backed by 300+ automated tests. Java 17, Spring Boot, Kafka, MySQL and AWS, delivered in Agile/Scrum teams.

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## TECHNICAL SKILLS

**Languages:** Java (11, 17, 21), Python, JavaScript, SQL, Deluge

**Backend Frameworks:** Spring Boot, Spring Cloud, Spring Security, Spring Data JPA, Hibernate, Spring MVC, Jersey, Servlets, JDBC, Undertow

**Architecture:** Microservices, Monolith Decomposition, REST API Design, Event-Driven Architecture, Webhook-Driven Services, RMI, Multi-Tenant SaaS

**Data & Messaging:** MySQL, PostgreSQL, SQLite, Cloudflare D1, Redis, Apache Kafka, Schema Migrations, Query Optimization

**AI & Agentic Systems:** Large Language Models (LLM), Multi-Agent Orchestration, Prompt Engineering, Model Context Protocol (MCP), Human-in-the-Loop AI

**Testing & Quality:** JUnit 5, Mockito, Test-Driven Development (TDD), Integration Testing, SAST Scanning, Secure Code Review

**Cloud & DevOps:** AWS (EC2, S3), Docker, Kubernetes, Cloudflare Workers and Pages, Gradle, Maven, Git, GitLab, CI/CD Pipelines, Deployment Automation

**Security & Integrations:** OAuth 2.0, HMAC-SHA256 Webhook Verification, Role-Based Access Control, Razorpay API, Meta WhatsApp Business API

**Practices:** Agile, Scrum, Cross-Functional Collaboration, Code Review, Application Monitoring and Analytics

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## PROFESSIONAL EXPERIENCE

### Member Technical Staff - Java Backend Engineer

Jul 2023 - Present

*Zoho Corporation - Customization Platform, Zoho Finance | Chennai / Coimbatore, India*

- Spearheaded the monolith-to-microservice migration of Zoho Books (Java 17), decoupling 15+ tightly coupled APIs into independently deployable services supporting 6+ Zoho Finance product lines including Zoho Books, Zoho Inventory, Zoho Expense, Zoho Payroll and Zoho Billing.
- Engineered a connector-based plug-and-play architecture that resolves cross-repository dependencies at startup, enabling each of the 6+ Finance products to build, deploy and release independently without coordinated downtime windows.
- Solved cross-service data dependencies with two complementary strategies: static metadata injected through HTTP headers at request initiation, and RMI for dynamically generated data inside the microservice layer.
- Built a real-time bidirectional sync engine between GoFrugal On-Premise (Vikra ONDC) and Zoho Finance applications using incremental sync and blob-based data processing; cleared internal security review and DPIA audit and is now in use by paying customers on the ONDC Network.
- Architected that sync framework around a generic connector pattern, letting the follow-on Zoho Commerce integration reuse the same extensible design and complete in 50% less time than the initial GoFrugal integration.
- Migrated 8+ core entity classes into a shared upper-level module consumed by 6+ Zoho Finance applications, coordinating a phased rollout across multiple product build cycles to avoid regressions.
- Designed and shipped the System Module Layout Customization feature for the Vendor and Sales Receipt modules, letting end users reorder and restructure field layouts; also owned the cross-product Shortcuts framework, integrating with internal configuration services to persist per-user keyboard settings across 6+ Finance products.
- Independently designed and built the Zoho Finance Developer Console in under one month, a JavaScript internal tool that unifies Gradle builds, server startup and sub-repository management into a visual workflow builder; adopted by 15+ engineers on the team.

- Identified and remediated 10+ security vulnerabilities, including a permission-level escalation in document upload workflows, through SAST scanning and manual code review ahead of each production deployment.
- Authored unit and integration tests (JUnit 5, Mockito) for migrated microservice modules, improving regression coverage across the shared platform layer.

## Member Technical Staff - Backend Engineer

Jun 2022 - Jul 2023

GoFrugal Technologies (Zoho group) - Connect Team, Alerts | Chennai, Tamil Nadu, India | Promoted from intern to full-time

- Integrated the Meta WhatsApp Business API into GoFrugal On-Premise and On-Cloud POS systems; the integration now ships as part of the GoFrugal product used by 1,000+ merchants across their network.
- Launched the Alert Gateway microservice on Undertow and Java 17 with Apache Kafka as the message broker, delivering asynchronous WhatsApp and SMS notifications from a single consolidated service.
- Built a Facebook template registration UI with emoji support, rich text formatting and live preview, enabling business users to create and manage WhatsApp notification templates at scale.
- Conducted security analysis across the codebase within Agile sprints, remediating 5+ critical vulnerabilities (SQL injection, path traversal, XSS) and hardening every module exposing open API endpoints.
- Managed application deployment on AWS EC2 and S3, supporting production infrastructure for Connect team services.

## PROJECTS

### Church Contribution Portal - Full-Stack Payments Platform

Dec 2024 - Present

Live in production | [light-of-jesus-ministry-contributions.pages.dev](https://light-of-jesus-ministry-contributions.pages.dev) | Cloudflare Pages Functions, D1, Razorpay

- Designed and shipped a live contribution management platform for a 40+ member community: 23 REST endpoints running as Cloudflare Pages Functions over a Cloudflare D1 database of 20 tables, evolved through 13 additive, backward-compatible migrations with zero production data loss.
- Automated the money path end to end with a Razorpay checkout plus a webhook receiver that verifies every callback with HMAC-SHA256 signature validation before recording a contribution, replacing manual spreadsheet reconciliation.
- Enforced role-based access control across 12 granular permissions and shipped a flag-gated v2 rollout behind a signed-cookie beta allowlist, so new flows reach opted-in users while production stays on the stable path.
- Hardened the platform with a 300+ case automated test suite that exercises the API handlers against a real in-memory SQLite instance, with CI blocking deployment on a red suite; migrated the original Google Apps Script backend onto Cloudflare with no downtime.

### Platform Copilot - AI Multi-Agent Workflow Automation

May 2026

Zoho Books Internal Hackathon

- Architected a two-agent LLM pipeline (Pattern Detector plus Executor) that reads Zoho Books activity logs, detects repetitive business patterns across 5 categories and auto-generates production-ready workflow automations with human-in-the-loop approval, cutting authoring time from roughly 30 minutes to about 30 seconds in prototype testing.
- Designed an enriched prefill JSON schema with a topological dependency map, enabling cross-agent coordination across 6 automation primitives: email templates, alerts, field updates, custom functions, webhooks and push notifications.
- Delivered a working production baseline, an active Zoho Books workflow with 3 wired actions (custom function, email alert, webhook) verified in a live org, and authored a hybrid MCP and UI playbook that cut manual workflow builds from 60 minutes to 5.
- Iterated the Executor system prompt across 3 versions against live API behaviour, diagnosing and documenting an MCP wrapper defect that silently dropped inline action objects, then rewriting the agent around a pre-create-then-reference pattern.

### TradeSync POS - Multi-Platform Business Management Suite

Oct 2024

[github.com/AlbertJoshwa1802011/allwin-traders-pos-main](https://github.com/AlbertJoshwa1802011/allwin-traders-pos-main)

- Developed a 3-platform business management suite comprising a voice-assisted Android delivery app, a Vercel-hosted web invoice manager and a customer-facing storefront, all synchronising in real time through Google Firebase on a multi-tenant database architecture.

## EDUCATION

### Bachelor of Engineering, Electronics and Communication Engineering

2018 - 2022

Sri Ramakrishna Engineering College, Coimbatore, Tamil Nadu | CGPA 8.1 / 10